

# STAT 8020 R Lab 8: Randomized Complete Block Designs and Factorial Designs

your name here, (the names of any collaborators)

## Contents

A researcher wishes to determine whether four different tips produce different hardness readings on a Rockwell hardness tester, which operates by pressing a tip into a metal test ‘coupon.’ Since the coupons are large enough to test four tips, a RCBD (Randomized Complete Block Design) can be used, with one coupon serving as a block. The dataset ‘tip\_hardness.csv’ can be found on Canvas.

1. Load the data into R.

**Code:**

2. Make side-by-side boxplots by treatment.

**Code:**

3. Write down the effects model and explain each term within the model, including the assumptions concerning the random error.

**Answer:**

4. Perform an overall  $F$ -test to the treatment effects using ANOVA, state the hypotheses,  $p$ -value, and conclusion.

**Code:**

**Answer:**

5. Use an interaction plot to assess the additive assumption.

**Code:**

**Answer:**

In a study to evaluate the impact of a certain bacterial strain (mycoplasma) on the development of birds, birds inoculated with bacteria or a control injection were studied in both cold and warm temperature-controlled rooms. The response variable of interest was the bill length of birds in the rooms. The rooms served as the experimental units, with either cold or warm temperatures, and contained either control or mycoplasma-inoculated birds. Two runs of each combination were performed. The data is given below:

| Bact       | Warm         | Cold         |
|------------|--------------|--------------|
| Control    | 40.37, 41.71 | 39.77, 40.23 |
| Mycoplasma | 40.21, 40.78 | 39.19, 38.95 |

6. Enter the data into R.

**Code:**

**Answer:**

7. Construct an ANOVA table that includes the main effects for bacteria and temperature and their interaction. Test whether the main effects impact bill length.

**Code:**

**Answer:**

8. Make an interaction plot and comment on the interaction effect.

**Code:**

**Answer:**